

Product Requirements Document (PRD)

Project Overview

A financial advisor marketing automation platform with AI-powered content generation and compliance workflows, specifically designed for FINRA/SEC-regulated financial advisors.

1 MVP Phase 1 – Features & Scope

Core Platform Features

Feature	Description	Tech / Stack
AI Content Generation (FINRA/SEC)	Advisors input prompt (e.g., "Roth conversions"). LLM generates compliant content for social, email, ads. Must incorporate FINRA/SEC rules.	AWS Lambda (TypeScript), OpenAI/Claude, Compliance prompt engineering
Content CRUD	Advisors create/edit content. Includes validation, versioning, and audit trail.	Vercel API Routes, PostgreSQL
Approval Workflow	Multi-step: Draft → Pending → Needs Changes → Approved → Posted. Compliance user can edit, approve, or request changes.	Vercel API, PostgreSQL
Compliance Gating	Content cannot be published until compliance-approved.	Role-based access, API checks
PDF/Word Export for Compliance	Generate PDF/Word of content for compliance review. Email to compliance team.	AWS Lambda, S3, SendGrid
Audit Trail	Track all content changes, approvals, and status history.	PostgreSQL
Email Templating & Sequencing	Create/send email campaigns, basic automation.	SendGrid API, Vercel API
Stripe Subscription & Billing	Subscription management, checkout, webhook handling.	Stripe API, Vercel API

LinkedIn Workstream

Feature	Description	Tech / Stack
Direct LinkedIn Posting	Post approved content to LinkedIn via official API.	LinkedIn API, AWS Lambda
LinkedIn Messaging Automation	Message sequencing and campaign automation via LinkedIn API.	AWS Lambda, LinkedIn API

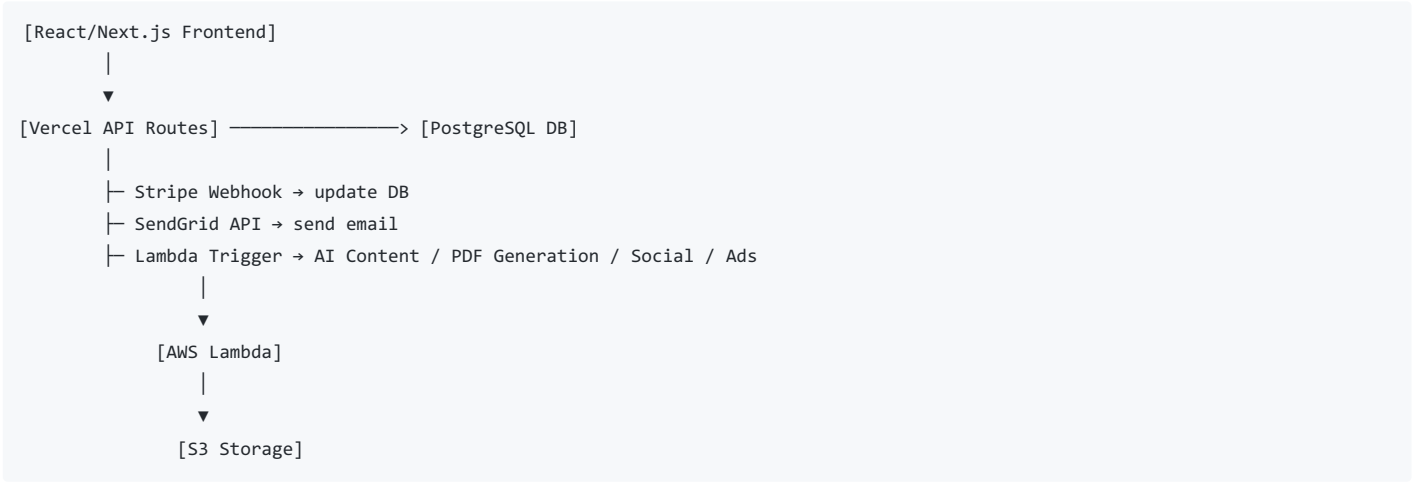
Facebook Workstream

Feature	Description	Tech / Stack
Direct Facebook Posting	Post approved content to Facebook via official API.	Facebook API, AWS Lambda
Facebook Ad Creation	Create/manage FB ads via Facebook Ad Form API.	AWS Lambda, FB Ad API

2 Tech Stack by Layer

Layer	Tech Stack	Responsibility
Frontend	React + Next.js + TypeScript + Tailwind	Dashboard, forms, tables, status display
Backend Light	Vercel API Routes	CRUD API, Stripe webhook, SendGrid API triggers
Backend Heavy	AWS Lambda (TypeScript)	AI content generation, PDF/Word export, social/ads
Database	PostgreSQL (Aurora Serverless / RDS)	Users, Content, Approvals, Audit Trail, API tokens
Storage	AWS S3	PDF / Word storage, download links
Auth	Clerk	User signup/login, role management, JWT
Integrations	OpenAI / Stripe / SendGrid / FB / LI	AI content, payment, emails, social, ads

3 Architecture Overview



- Frontend triggers Vercel serverless functions for CRUD, webhook handling, email sending.
- For heavy tasks (AI, PDF/Word, social, ads), Vercel functions call AWS Lambda directly.
- Generated PDFs/Words are stored in S3.
- All content + approvals stored in PostgreSQL.
- Social/Ad posting via Lambda integrations.

4 Estimations per Stack

Frontend (React + Next.js + TypeScript + Tailwind)

Task	Estimation (days)	Notes
Auth Integration (Clerk)	2	Signup, login, roles
Dashboard / Tables / Forms	6	Content list, status view, inline comments
Approval Workflow UI	3	Status buttons, modals, comments
Stripe Pricing Page	2	Checkout integration
Email / Notification UI	1	Basic email triggers
LinkedIn Posting UI	1	LI post interface
Facebook Posting/Ad UI	1	FB post, ad forms

Task	Estimation (days)	Notes
Total Frontend	18 days	

Backend Light (Vercel API Routes / TypeScript)

Task	Estimation (days)	Notes
CRUD API (Content)	5	Create / read / update / delete + validation
Approval Workflow Logic	4	Status changes, rules enforcement
Stripe Webhook	2	Idempotency + DB sync
SendGrid Email API	3	Trigger emails on approve
Audit Trail Logging	2	Save all changes / versioning
LinkedIn API Triggers	1	Call Lambda for LI
Facebook/Ad API Triggers	1	Call Lambda for FB/Ad
Total Backend Light	18 days	

Backend Heavy (AWS Lambda + TypeScript)

Task	Estimation (days)	Notes
AI Content Generation (<60 sec)	5	Lambda function + OpenAI API call
PDF / Word Generation	5	Puppeteer / PDFKit, S3 upload
LinkedIn Posting Integration	2	LI API, OAuth, error handling
Facebook Posting Integration	1	FB API, OAuth, error handling
Ad Creation Integrations	3	FB Ad API, campaign logic
Direct Lambda Calls from Vercel	1	Integration testing, error handling
Total Backend Heavy	17 days	

Database & Storage

Task	Estimation (days)	Notes
PostgreSQL Schema	2	Users, Content, Approvals, Audit
S3 Setup	1	Bucket policies, access control
Total DB & Storage	3 days	

Security / Auth / Logging

Task	Estimation (days)	Notes
Clerk auth integration	1	Already partially in frontend
JWT / Role enforcement	2	Backend API checks

Audit log enforcement Task	1 Estimation (days)	Confirm all changes saved Notes
Total Security	4 days	

Overall Stage 1 Estimation

Layer	Days
Frontend	16
Backend Light	18
Backend Heavy	17
DB & Storage	3
Security & Auth	4
Total	58 days (~3 months for 1 dev)

Notes: QA / buffer +5–7 days; team of 2 → ~1.5–2 months

5 Key Points & Out-of-Scope (Stage 1)

- All code in TypeScript monorepo (frontend + backend + lambda functions)
- No SQS/EventBridge in Stage 1 (direct Lambda calls)
- S3 is used for all PDF/Word exports
- Social/Ad automation is basic (no advanced retry/queueing)
- Compliance workflow is enforced at all publishing points
- All integrations (FB, LI, Stripe, SendGrid) are direct, with OAuth where required
- Can scale later by introducing queues, job scheduling, retry logic

6 Optional / Deferred Features (Stage 2+)

Feature	Notes
LinkedIn Messaging Sequences	Requires queueing / rate-limit handling → SQS later
Facebook Ads Automation	Campaign → Ad Set → Creative → Lead Form → requires retry / status checking
Advanced Retry Logic / Idempotency	For heavy automation / scheduled jobs
Job Scheduling / Cron	For recurring posting / email
Instagram Posting	API integration, similar to FB/LI
Analytics Dashboard	Impressions, connections, campaign stats
Calendar View	Scheduling, drag-and-drop UI

7 Estimations for Stage 2

Frontend (React + Next.js + TypeScript + Tailwind)

Task	Estimation (days)	Notes

Analytics Dashboard Task	4	Estimation (days) Charts, metrics, data fetching Notes
Calendar View	3	Drag-and-drop scheduling UI
Total Frontend	7 days	

Backend Light (Vercel API Routes / TypeScript)

Task	Estimation (days)	Notes
API Updates for Analytics	2	Endpoints for metrics, calendar data
Total Backend Light	2 days	

Backend Heavy (AWS Lambda + TypeScript)

Task	Estimation (days)	Notes
LinkedIn Messaging Sequences	5	SQS integration, rate limiting
Facebook Ads Automation	5	Advanced campaign logic, retry
Advanced Retry / Idempotency	3	Across all automations
Job Scheduling / Cron	3	EventBridge setup, recurring jobs
Instagram Posting	2	API integration, similar to FB/LI
Total Backend Heavy	18 days	

Database & Storage

Task	Estimation (days)	Notes
Schema Updates	1	Analytics tables, scheduling
Total DB & Storage	1 day	

Security / Auth / Logging

Task	Estimation (days)	Notes
Updates for New Features	1	Ensure compliance
Total Security	1 day	

Overall Stage 2 Estimation

Layer	Days
Frontend	7
Backend Light	2
Backend Heavy	18
DB & Storage	1

Security & Auth Layer	1 Days
Total	29 days (~1.5 months for 1 dev)

Notes: QA / buffer +3–5 days; team of 2 → ~1 month

8 UI Components (MVP)

- Dashboard with analytics (basic)
- Multi-platform posting interface (LinkedIn, Facebook)
- Campaign/ad creation forms
- Calendar view (basic, optional)
- Approval workflow modals and status badges
- Email/campaign template editor

9 Compliance & Security Notes

- LLM prompts must embed FINRA/SEC rules
- All publishing is gated by compliance approval
- Audit trail is mandatory for all content changes
- PDF/Word export for compliance review is required
- Role-based access for Advisors, Compliance, Admin

10 Summary

This PRD covers all MVP Phase 1 requirements, tech stack, and realistic estimates. It ensures:

- FINRA/SEC-compliant AI content
- End-to-end compliance workflow
- Direct social/email/ad integrations
- Secure, auditable, and scalable foundation